

# HDOC Day-13 Solution

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## Menu-Driven C Program using switch-case

### 1. Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one operation based on the user's choice:

- Choice 1: Count the number of even digits and odd digits. Example: 12045 → Even digits = 3, Odd digits = 2.
- Choice 2: Find the sum of even digits and the sum of odd digits separately. Example: 12345 → Sum of even digits = 6, Sum of odd digits = 9.

### 2. Algorithm

1. Start.
2. Input a positive integer n.
3. Display the menu: 1) Count even and odd digits, 2) Find sum of even and odd digits.
4. Input the user's choice.
5. Use switch(choice) to select the required operation.
6. For Choice 1: set even\_count = 0 and odd\_count = 0. Examine each digit using digit = n % 10. If digit % 2 == 0, increment even\_count; otherwise increment odd\_count. Remove the digit using n = n / 10. Display both counts.
7. For Choice 2: set sum\_even = 0 and sum\_odd = 0. Examine each digit. Add an even digit to sum\_even and an odd digit to sum\_odd. Display both sums.
8. For any other choice, display "Invalid choice".
9. Stop.

### 3. Test Case Table

The table below tests both menu operations with easy-to-check inputs.

| Test | Operation             | Input number | Choice | Expected output                                  | Purpose                                 |
|------|-----------------------|--------------|--------|--------------------------------------------------|-----------------------------------------|
| TC1  | Count even/odd digits | 12045        | 1      | Even digits = 3<br>Odd digits = 2                | Given example; includes digit 0 (even). |
| TC2  | Count even/odd digits | 2468         | 1      | Even digits = 4<br>Odd digits = 0                | Checks an all-even number.              |
| TC3  | Count even/odd digits | 13579        | 1      | Even digits = 0<br>Odd digits = 5                | Checks an all-odd number.               |
| TC4  | Sum even/odd digits   | 12345        | 2      | Sum of even digits = 6<br>Sum of odd digits = 9  | Given example.                          |
| TC5  | Sum even/odd digits   | 2468         | 2      | Sum of even digits = 20<br>Sum of odd digits = 0 | Checks sums for all-even digits.        |
| TC6  | Sum even/odd digits   | 13579        | 2      | Sum of even digits = 0<br>Sum of odd digits = 25 | Checks sums for all-odd digits.         |

### 4. C Program

```
#include <stdio.h>

int main()
{
    int n, temp, digit, choice;
    int even_count = 0, odd_count = 0;
    int sum_even = 0, sum_odd = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\n1. Count even and odd digits\n");
    printf("2. Find sum of even and odd digits\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = n;

    switch (choice)
    {
        case 1:
            while (temp > 0)
            {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_count++;
                else
                    odd_count++;
                temp = temp / 10;
            }
            printf("Even digits = %d\n", even_count);
            printf("Odd digits = %d\n", odd_count);
            break;
```

```

        case 2:
            while (temp > 0)
            {
                digit = temp % 10;
                if (digit % 2 == 0)
                    sum_even = sum_even + digit;
                else
                    sum_odd = sum_odd + digit;
                temp = temp / 10;
            }
            printf("Sum of even digits = %d\n", sum_even);
            printf("Sum of odd digits = %d\n", sum_odd);
            break;

        default:
            printf("Invalid choice.\n");
    }

    return 0;
}

```

## 5. Sample Execution

### Choice 1

```

Enter a positive integer: 12045
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 3
Odd digits = 2

```

### Choice 2

```

Enter a positive integer: 12345
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum of even digits = 6
Sum of odd digits = 9

```

## 6. Screenshot Space

### Screenshot 1 – Source code / compilation

```

#include <stdio.h>
int main()
{
    int n, temp, digit, choice;
    int even_count = 0, odd_count = 0;
    int sum_even = 0, sum_odd = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }
    printf("\n1. Count even and odd digits\n");
    printf("2. Find sum of even and odd digits\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    temp = n;
    switch (choice)
    {
        case 1:
            while (temp > 0)
            {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_count++;
                else
                    odd_count++;
                temp = temp / 10;

```

```

            temp = temp / 10;
        }
        printf("Even digits = %d\n", even_count);
        printf("Odd digits = %d\n", odd_count);
        break;
        case 2:
            while (temp > 0)
            {
                digit = temp % 10;
                if (digit % 2 == 0)
                    sum_even = sum_even + digit;
                else
                    sum_odd = sum_odd + digit;
                temp = temp / 10;
            }
            printf("Sum of even digits = %d\n", sum_even);
            printf("Sum of odd digits = %d\n", sum_odd);
            break;
        default:
            printf("Invalid choice.\n");
    }
    return 0;
}

```

## Screenshot 2 – Execution for Choice 1

```

(kali@kali)-[~/Desktop]
$ ./sample9
Enter a positive integer: 12045

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 3
Odd digits = 2

```

```

(kali@kali)-[~/Desktop]
$ ./sample9
Enter a positive integer: 2468

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 4
Odd digits = 0

```

```
(kali㉿kali)-[~/Desktop]
$ ./sample9
Enter a positive integer: 13579

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 0
Odd digits = 5
```

### Screenshot 3 – Execution for Choice 2

```
(kali㉿kali)-[~/Desktop]
$ ./sample9
Enter a positive integer: 12345

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum of even digits = 6
Sum of odd digits = 9
```

```
(kali㉿kali)-[~/Desktop]
$ ./sample9
Enter a positive integer: 2468

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum of even digits = 20
Sum of odd digits = 0
```

```
(kali㉿kali)-[~/Desktop]
$ ./sample9
Enter a positive integer: 13579

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum of even digits = 0
Sum of odd digits = 25
```